Vaibhav Panchal

Junior AI/ML Developer | GenAI & LLM Systems Engineer
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PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs Inc. building intelligent test automation pipelines using deep learning and LLMs. M.E. in Software Engineering (LDCE, GTU, CGPA 8.21/10). Research work in Computer Vision (YOLOv11 + ECA, +3.8% mAP50 on DIOR-R, under review) and Cybersecurity (hybrid IDS, 98.4% accuracy, unpublished manuscript). Hands-on across the full ML lifecycle from data engineering and model training to Flask deployment and CI/CD integration. Core stack: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, Flask, Nginx, AWS, Docker.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, JavaScript, C/C++, Java, HTML/CSS
Frameworks and Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, XGBoost, LangChain, LangGraph, FastAPI, Flask, React.js, Node.js, Streamlit
AI and ML: Computer Vision, NLP, LLMs, RAG, Object Detection, Prompt Engineering, OpenAI API, Ollama, RAGAS, Langfuse, NeMo Guardrails
Cloud and Infrastructure: AWS (EC2), Docker, Nginx
Databases: SQLite, MongoDB, ChromaDB, Pinecone (Vector DB)
Tools and Practices: Git, Jupyter, MLflow, Postman, Selenium Grid, Bitbucket, REST API Design, CI/CD

WORK EXPERIENCE

Junior AI/ML Developer | TestGrid Labs Inc. | Navsari, Gujarat *May 2026 - Present*

- Engineered deep learning inference pipelines for automated UI regression detection, reducing manual test triage time by flagging anomalies before human review.
- Designed and deployed Flask-based model-serving endpoints integrated into the CI/CD pipeline, enabling real-time ML inference on test runs at scale.
- Prototyped an LLM-assisted test case generation module using the OpenAI API, reducing repetitive test authoring effort across the QA team.
- Configured Nginx as a reverse proxy in front of Flask model-serving endpoints, handling request routing and load distribution for production ML inference traffic.

Trainee Associate Software Developer | Techreale Pvt. Ltd. | Valsad, Gujarat *Jan 2024 - Apr 2024*

- Delivered production features for the Famyll healthcare platform (React.js, Node.js, MongoDB) across claim management, auth flows, and real-time communication modules.
- Reduced API response handling bugs by refactoring async integration patterns; contributed across 3 sprint cycles in an Agile team of 8.

Summer Intern, Frontend Development (React) | INFOLABZ IT Services Pvt. Ltd. | Ahmedabad, Gujarat *Jul 2023*

- Developed and optimized React.js dashboard components with async API integration; improved rendering performance through targeted component-level refactoring.

PROJECTS

MESH Support - Multi-Agent Customer Support Platform *2026*
Python | LangGraph | FastAPI | MongoDB | GPT-5.4-mini | NeMo Guardrails | Langfuse | Pytest

- Architected MESH Support using LangGraph to orchestrate a Supervisor Agent that classifies and routes queries to specialized Billing, Technical, and Refund sub-agents, each with isolated prompts, tools, and memory, backed by FastAPI, MongoDB, and GPT-5.4-mini.
- Integrated NVIDIA NeMo Guardrails input/output rails to block jailbreak, prompt-injection, off-topic, and unsafe messages, achieving 100% precision/recall (0% false-positive rate) on a labeled safety set with only ~1.8s added overhead per turn.
- Instrumented every turn with Langfuse distributed tracing and shipped a real-time observability dashboard visualizing agent hops and tool calls; verified reliability with an automated pytest suite (21/21 tests passing).
- Delivered 100% end-to-end accuracy on the evaluation suite (routing, tool-selection, and guardrail decisions) with a ~5.4s median guarded response time, validated by automated end-to-end test and evaluation harnesses (23/23 backend tests passing).

RAG System with Evaluation Pipeline *2026*
Python | FastAPI | React | Pinecone | Langfuse | RAGAS | MiniLM | BM25 | NeMo Guardrails

- Achieved 0.94 Faithfulness, 0.91 Answer Relevancy, and 0.88 Context Precision (RAGAS) on a golden test set of domain-specific policy and technical Q&A pairs, ensuring highly context-grounded responses and minimizing hallucinations.
- Built hybrid retrieval (dense MiniLM-384d embeddings + BM25 keyword search via Reciprocal Rank Fusion) achieving ~920ms average end-to-end query latency, with a BM25 auto-fallback (<15ms in-memory rebuild) for zero-downtime operation when the vector store is unreachable.
- Enforced input safety with NeMo Guardrails, blocking 100% of tested prompt injections, jailbreak attempts, and off-topic queries before they reached the retrieval pipeline.
- Instrumented 100% trace coverage with Langfuse across every retrieval and generation span, enabling per-metric scoring and side-by-side A/B comparison of retrieval strategies.

ExamForge - LLM-Powered Exam Question Generator *2026*
Python | LangChain | Ollama | OpenAI GPT | Streamlit
- Built an end-to-end RAG pipeline: ingests syllabus PDFs, chunks and embeds content, then prompts an LLM to generate structured question papers with configurable difficulty and Bloom's taxonomy tagging.
- Supported both local inference (Ollama) and cloud (OpenAI API), making it deployable in air-gapped institutional environments; cut manual question-drafting time by ~80% in pilot use.
- Engineered structured JSON output parsing with validation to ensure generated questions conform to a strict schema before rendering to PDF.

Content-Based Movie Recommendation Engine *Jan - Feb 2025*
Python | Scikit-learn | TF-IDF | Cosine Similarity | Streamlit
- Implemented TF-IDF vectorization and cosine similarity on the TMDB 5000 dataset; deployed as a real-time Streamlit app with sub-second recommendation latency.
- Pipeline is architected for extensibility - collaborative filtering and hybrid approaches can be plugged in without restructuring the serving layer.

RESEARCH WORK

Channel Attention for Oriented Object Detection in Remote Sensing *2025 (Under Review)*
YOLOv11-OBB | ECA Attention | DIOR-R Dataset | PyTorch
- Integrated Efficient Channel Attention (ECA) into YOLOv11s-OBB to improve feature recalibration for small and densely packed objects in aerial imagery.
- Evaluated on DIOR-R (20-class remote sensing benchmark): achieved +3.8% mAP50 and +4.3% recall over the baseline, with no increase in parameter count.

Hybrid Deep Learning Architecture for Network Intrusion Detection *2025 (Manuscript in Preparation)*
CNN + BiLSTM + Attention | NSL-KDD | CICIDS | TensorFlow
- Proposed a sequential CNN-BiLSTM-Attention IDS that captures both spatial and temporal patterns in network flow data; applied PCA and AutoEncoder for feature compression.
- Achieved 98.4%+ accuracy on NSL-KDD and CICIDS; outperformed Random Forest and SVM across all attack categories on precision, recall, and F1-score.

EDUCATION

M.E. - Software Engineering - GTU, LDCE, Ahmedabad 2024-2026 | CGPA: 8.21/10
B.E. - Computer Engineering - GTU, GDEC, Navsari 2020-2024 | CGPA: 8.11/10

ACHIEVEMENTS & CERTIFICATIONS

- Evaluate Generative AI Applications - Microsoft Learn; covered evaluation frameworks and metrics for Azure, Foundry Tools, and Content Safety. (2026)
- Microsoft Learn Challenge | Ignite Edition: Build Trustworthy AI Solutions on Microsoft Azure. (2025)
- SSIP Hackathon 2022 - Developed an automated smart attendance management system; selected among top teams from Gujarat. (2022)
- Dev-o-lution Hackathon - Certificate of Participation for building a competitive solution under a 24-hour constraint. (2025)
- Introduction to Model Context Protocol – Anthropic. (2026)
- Model Context Protocol: Advanced Topics – Anthropic. (2026)

LEADERSHIP & EXTRACURRICULAR

- Volunteered at college-level tech events and coding quiz competitions, supporting event coordination and participant engagement.
- Academic Mentorship - Mentored B.E. students in computer science subjects during my M.E., providing guidance on technical concepts and problem-solving.

INTERESTS

AI/ML Research, LLMs & RAG Systems, Computer Vision, Deep Learning, MLOps, Automation, Cloud Computing, Data Science | Chess, Exploring Emerging Technologies, Music, Video Games